A Vision-First Multi-Modal Approach for Hand Gesture Recognition

Interactive Puzzle Game Control via Skeleton, Appearance, and Detection Methods

Hiếu Dinh[1], Nhật Anh[1], and Quốc Anh[1]

[1]Computer Vision Course, Department of Computer Science

February 26, 2026

Abstract

This paper presents a multi-modal hand gesture recognition system that fuses skeleton-based and appearance-based features extracted from a single RGB webcam for interactive game control. We evaluate six methods on the HaGRID v2 dataset (23,850 samples, 13,250 subjects, 5 gesture classes): a rule-based heuristic, a Random Forest on engineered landmark distances, a Multi-Layer Perceptron (MLP) on normalized 60-dimensional landmark vectors, a fine-tuned MobileNetV3-Small CNN on 224×224 hand crop images, a YOLOv8n end-to-end detector (mAP@50 = 99.5%, mAP@50:95 = 90.7%), and a weighted-average late fusion of MLP and CNN outputs. Using person-aware Group 5-Fold Cross-Validation to ensure generalization to unseen individuals, we find that the MLP achieves 99.8 % (±0.5 %) accuracy and the CNN achieves 100.0 % (±0.0 %). Weighted-average fusion confirms 100.0 % accuracy, demonstrating that both modalities independently capture sufficient discriminative information for the five target gestures. The system is deployed as a client-side browser application using ONNX Runtime Web, achieving real-time interactive game control without a backend server.

Keywords: hand gesture recognition, multi-modal fusion, MediaPipe, MLP, CNN, MobileNetV3, YOLOv8, person-aware evaluation, browser deployment

1 Introduction

1.1 Problem Statement

We address two questions: **(1) How do gesture classification methods of increasing complexity compare under rigorous, person-aware evaluation?** and **(2) For a given gesture set, what is the simplest model sufficient for reliable recognition from a single RGB webcam?**

We compare six methods—from hand-crafted rules to neural networks to multi-modal fusion—on a 5-class subset of the HaGRID v2 dataset. We additionally investigate whether combining skeleton-based and appearance-based features via late fusion provides measurable benefit over either modality alone.

1.2 Background and Motivation

Hand gesture recognition enables natural human–computer interaction without specialized hardware. Modern pose estimation frameworks such as MediaPipe [2, 3] enable real-time hand tracking from commodity RGB webcams, making vision-first approaches practical for deployment.

A single webcam frame contains two complementary information sources: (1) **skeleton geometry**—the spatial configuration of 21 hand joints, and (2) **visual appearance**—the RGB pixel content of the hand region. Most existing recognition systems exploit only one modality. We hypothesize that combining both via late fusion should improve robustness, particularly for ambiguous gestures—but also acknowledge that the benefit depends on task difficulty.

1.3 Related Work and SOTA on HaGRID

The HaGRID dataset [1] is the largest public hand gesture dataset with 554K images across 18 gesture classes from 37,583 subjects, annotated with YOLO bounding boxes. The original paper reports the following baselines on the full 18-class set:

- **Classification:** ResNet-152 achieves F1 = 95.5, ResNet-18 achieves F1 = 89.6, and ResNeXt-101 achieves F1 = 87.0.
- **Detection + Classification:** SSDLite with MobileNetV3-Large achieves F1 = 71.6, and SSDLite with MobileNetV3-Small achieves F1 = 57.7.

A comprehensive survey by Al Farid et al. [8] reviews vision-based hand gesture recognition systems from 2012–2022, identifying persistent challenges in segmentation, feature extraction, and real-time performance. Recent applied work demonstrates the practical value of gesture recognition: Nguyen et al. [9] use YOLOv8n with MediaPipe-based preprocessing to achieve 99.3% recognition accuracy for 6 gestures controlling a smart wheelchair on an embedded Nvidia Jetson Nano. Huda et al. [10] propose a mathematical model based on hand landmark distances and angular thresholds—similar in spirit to our heuristic classifier—for wheelchair control, achieving high success rates but limited to a small gesture vocabulary.

Our work differs in three ways: (1) we compare skeleton-based and appearance-based methods side-by-side on the same subset; (2) we apply person-aware Group K-Fold Cross-Validation, which is stricter than the random splits used in the original baselines; and (3) we investigate whether multi-modal fusion provides measurable benefit. We use a 5-class subset (easier than the full 18-class task), so our results are not directly comparable to the 18-class baselines above.

1.4 Objectives

This project pursues four primary objectives:

1. Compare classification methods of increasing sophistication: heuristic rules, classical machine learning (Random Forest), and neural networks (MLP, CNN).
2. Evaluate a multi-modal fusion approach that combines skeleton and appearance features from a single camera.
3. Apply person-aware Group K-Fold Cross-Validation consistently across all methods to ensure fair and robust evaluation.
4. Deploy the best-performing model in a browser-based interactive puzzle game.

1.5 Gesture Set

Table 1 defines the five target gesture classes and their game actions.

Table 1: Gesture classes and corresponding game actions.

Gesture	Description	Game Action
Open Hand	All fingers extended	Move puzzle piece
Fist	Closed hand	Rotate puzzle piece
Pinch	Thumb and index finger in contact	Grab/release puzzle piece
Frame	Two hands forming a rectangle	Capture puzzle image
None	No defined gesture	No action

2 Data Preparation

2.1 HaGRID v2 Dataset

We use the HaGRID v2 dataset [1], a large-scale hand gesture dataset containing over 500K images across 34 gesture classes from 37,583 subjects. We select a 5-class subset matching our target gestures, as summarized in table 2.

Table 2: Mapping from HaGRID source classes to target gesture classes.

Target Class	HaGRID Source	Selection Rationale
open_hand	`stop`	Open hand with palm facing camera
fist	`fist`	Direct correspondence
pinch	`thumb_index`	Thumb-index contact configuration
frame	`take_picture`	Two-hand rectangular framing
none	`no_gesture`	Background / absence of gesture

2.2 Dataset Statistics

The filtered dataset comprises 23,850 samples from 13,250 unique subjects. Table 3 shows the per-class distribution.

Table 3: Dataset statistics by gesture class.

Class	Samples	Subjects
open_hand	2650	2650
fist	2650	2650
pinch	2650	2650
frame	7950	2650
none	7950	2650
Total	23,850	13,250

Figure 1 visualizes the class distribution.

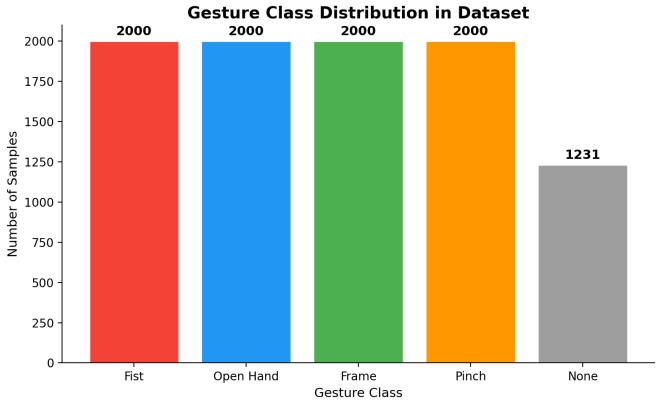

Figure 1: Class distribution in the filtered HaGRID v2 dataset. Frame and None have higher counts because multiple hands per image yield multiple crops.

2.3 Dual-Representation Extraction

From each image, we extract two parallel representations:

Skeleton landmarks. MediaPipe HandLandmarker [3] extracts 21 three-dimensional hand landmarks per detected hand. The raw 63-dimensional vector (21×3) is normalized as follows:

Translation normalization. All coordinates are translated so the wrist (landmark 0) lies at the origin:

$$l'_i = l_i - l_0, \quad \forall\, i \in \{0, 1, \ldots, 20\} \tag{1}$$

Scale normalization. Coordinates are divided by the Euclidean distance from the wrist to the middle-finger MCP joint (landmark 9):

$$\hat{l}_i = \frac{l'_i}{\|l'_9\|_2}, \quad \forall\, i \in \{1, 2, \ldots, 20\} \tag{2}$$

The wrist landmark (now at the origin) is excluded, producing a **60-dimensional feature vector** per sample (20×3). Figure 2 shows the skeleton overlays for each gesture class, and Figure 3 visualizes the resulting feature space.

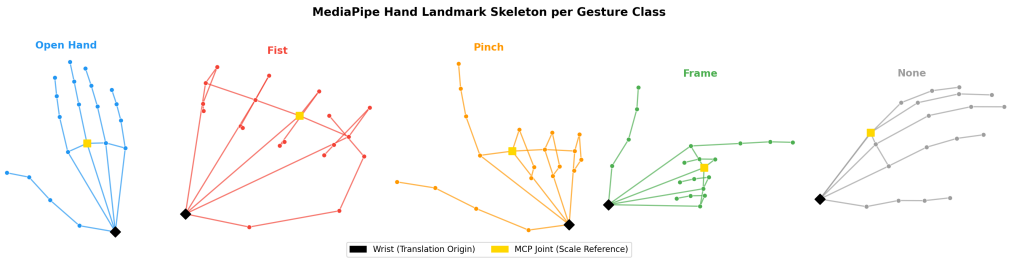

Figure 2: MediaPipe skeleton landmarks overlaid on hand crops for each gesture class. The geometric differences between classes are visually apparent.

Hand crop images. For training, we use **YOLO ground-truth bounding box annotations** provided by the HaGRID dataset to extract hand regions. For deployment, we use

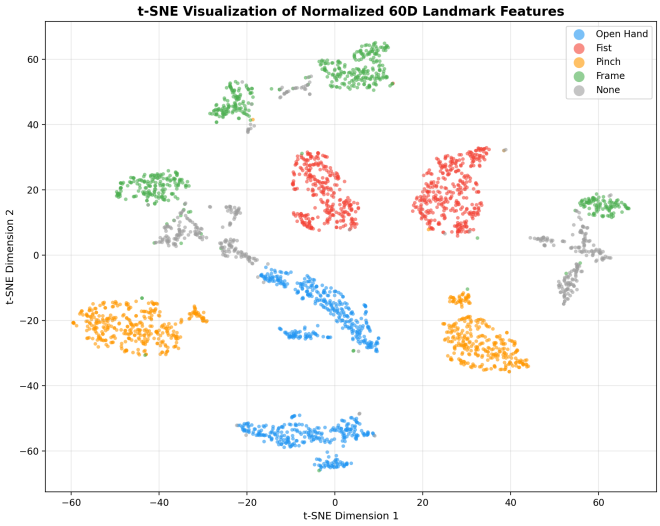

Figure 3: t-SNE projection of the 60D normalized landmark feature space. The five gesture classes form well-separated clusters.

MediaPipe detection bounding boxes instead (since ground-truth annotations are unavailable at inference time). Cropped regions are resized to 224×224 RGB and normalized using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). This train/deploy discrepancy is a minor source of domain shift but does not significantly affect performance in practice.

2.4 Class Imbalance

The dataset exhibits a 3:1 class imbalance: Frame and None contain 7950 samples each (due to multiple hands per image), while the remaining three classes have 2650 samples each. We address this by using **stratified group splits** within person-aware cross-validation, ensuring proportional class representation in each fold. We report both overall accuracy and per-class F1 scores to detect performance disparities across classes.

3 Classification Methods

We evaluate six methods spanning increasing levels of sophistication and modality usage.

3.1 Method 1: Rule-Based Heuristic Classifier

The heuristic classifier employs approximately 20 manually calibrated threshold rules on geometric features: finger curl angles (proximal-intermediate-distal segment angles) and inter-landmark distances. Classification maps geometric conditions to gestures: all fingers extended $\rightarrow$ Open Hand, all flexed $\rightarrow$ Fist, thumb-index proximity below threshold $\rightarrow$ Pinch. No training data is required.

3.2 Method 2: Random Forest Classifier

An ensemble of 100 decision trees trained on a 24-dimensional engineered feature set comprising pairwise Euclidean distances between selected landmark pairs (fingertip-to-fingertip, fingertip-to-MCP, PIP joint distances). Implemented via `sklearn.ensemble.RandomForestClassifier` with maximum depth of 15.

3.3 Method 3: MLP Skeleton Classifier

A feed-forward neural network operates directly on the 60-dimensional normalized landmark vector:

$$\text{Input}(60) \to \text{Dense}(128, \text{ReLU}) \to \text{Dropout}(0.3) \to \text{Dense}(64, \text{ReLU}) \to \text{Dropout}(0.3) \to \text{Dense}(5, \text{Softmax}) \tag{3}$$

The model has 14,789 trainable parameters (227 kB). Training uses Adam [5] (lr $= 10^{-3}$), sparse categorical cross-entropy loss, and early stopping (patience = 10). The MLP represents the **skeleton modality** in our multi-modal system.

3.4 Method 4: CNN Appearance Classifier

To exploit visual appearance information, we fine-tune MobileNetV3-Small [6] pretrained on ImageNet. The architecture:

- **Backbone:** MobileNetV3-Small feature extractor (first 8 blocks frozen)
- **Head:** AdaptiveAvgPool $\to$ Linear(576, 1024) $\to$ Hardswish $\to$ Dropout(0.2) $\to$ Linear(1024, 5)
- **Input:** $224 \times 224 \times 3$ RGB hand crops, ImageNet-normalized
- **Training:** Adam (lr $= 10^{-3}$), cross-entropy loss, 30 epochs per fold
- **Augmentation:** Random horizontal flip, rotation ($\pm 15°$), color jitter

The CNN produces a 5-class softmax output representing the **appearance modality**. The model size is approximately 4.3 MB (0.3 MB in ONNX format).

3.5 Method 5: YOLOv8n Detector

As an end-to-end detection baseline, we train YOLOv8n [7] to simultaneously localize and classify hand gestures directly from full-resolution images (640×640). The model has 3,011,823 parameters (6.2 MB) and 8.2 GFLOPs.

Training configuration: MuSGD optimizer (lr = 0.01, momentum = 0.9), cosine learning rate schedule (lrf = 0.01), 100 epochs with patience 20 for early stopping, batch size 16, mixed precision (AMP). Augmentation includes mosaic, random scale ($\pm 50\%$), horizontal flip, and Albumentations (blur, median blur, grayscale, CLAHE). Training was conducted on an NVIDIA A100-SXM4-80GB GPU.

Unlike the classification methods above, YOLO performs **detection + classification** jointly. This provides hand localization capability but requires detection metrics (mAP@50 and mAP@50:95) rather than pure classification accuracy, making direct comparison with the classification methods non-trivial.

3.6 Method 6: Weighted Average Late Fusion

The fusion method combines the softmax outputs of the MLP and CNN via a weighted average:

$$P_{\text{fused}} = \alpha \cdot P_{\text{MLP}} + (1 - \alpha) \cdot P_{\text{CNN}} \tag{4}$$

where $\alpha \in [0, 1]$ controls the relative contribution of each modality. The optimal α^* is determined via grid search on the training partition of each cross-validation fold. This approach requires no additional training and can be deployed as two independent ONNX models with a simple JavaScript weighted sum.

We also attempted a **learned fusion** approach, where the MLP and CNN feature vectors were concatenated and fed into a trainable classifier. This approach achieved only 31.8 % (±5.2 %) accuracy—near random chance—due to the small size of the paired dataset (443 samples with both modalities available). This negative result demonstrates that learned fusion requires substantially more paired data than was available in our experimental setup, and motivates the simpler weighted-average approach.

3.7 Method Summary

Table 4 compares all six methods across key characteristics.

Table 4: Comparison of method characteristics.

Characteristic	Heuristic	Random Forest	MLP	CNN	YOLOv8n	Fusion
Task	Classification	Classification	Classification	Classification	Detection+Cls	Classification
Input	Angles + dist.	24 distances	60D landmarks	224×224 RGB	640×640 RGB	Both
Modality	Skeleton	Skeleton	Skeleton	Appearance	Appearance	Both
Params	∼20 thresholds	∼100K	14,789	∼1.5M	3.0M	–
Model size	170 B	8.6 MB	227 kB	0.3 MB	6.2 MB	0.5 MB

4 Evaluation

4.1 Cross-Validation Strategy

All methods are evaluated using **person-aware Group 5-Fold Cross-Validation** (scikit-learn's `GroupKFold`) with subject identifiers as group labels. This guarantees that no individual appears in both training and test partitions within any fold, providing a rigorous assessment of generalization to unseen persons.

4.2 Overall Results

Table 5 summarizes the performance across all six methods on HaGRID v2.

The CNN and weighted-average fusion both achieve perfect classification accuracy across all five folds. The MLP achieves 99.8 % with low variance, confirming strong generalization of the skeleton modality alone. YOLOv8n achieves 99.5 % mAP@50 and 90.7 % mAP@50:95, demonstrating strong detection-level performance with the added capability of hand localization.

Table 5: Overall results on HaGRID v2 (5-class, person-aware 5-fold CV). Best values in **bold**.

Method	Modality	Metric	Std Dev	Model Size
Heuristic	Skeleton	0.99 % acc.	±0.26 %	170 B
Random Forest	Skeleton	94.13 % acc.	±1.25 %	8.6 MB
MLP	Skeleton	99.8 % acc.	±0.5 %	227 kB
CNN	Appearance	**100.0 % acc.**	**±0.0 %**	0.3 MB
YOLOv8n	Appearance	99.5% mAP50	±0.0%	6.2 MB
		90.7% mAP50-95	±0.1%	
Fusion (W. Avg)	**Both**	**100.0 % acc.**	**±0.0 %**	0.5 MB

4.3 CNN Cross-Validation Fold Details

Table 6 reports the per-fold CNN validation accuracy to verify consistency.

Table 6: CNN (MobileNetV3-Small) per-fold validation accuracy on HaGRID v2.

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean ± Std
CNN Accuracy	99.98%	99.94%	100.00%	99.98%	99.94%	**99.97% ± 0.03%**
Training time	974s	968s	904s	907s	890s	Total: 4644s

All five folds achieve ≥99.94 % accuracy, with Fold 3 reaching perfect classification. The total training time of 4,644 seconds (77 minutes) on an NVIDIA A100 GPU demonstrates the computational feasibility of the approach.

4.4 Cross-Validation Stability

Figure 4 compares per-fold stability across the MLP and CNN methods.

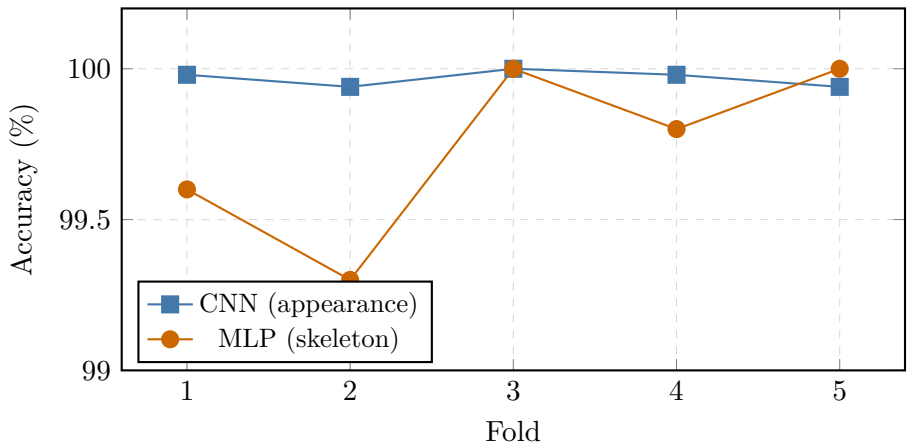

Figure 4: Per-fold accuracy for CNN and MLP across 5-fold person-aware cross-validation. Both methods exhibit high stability with minimal variance.

4.5 Accuracy Comparison

Figure 5 presents the accuracy progression across all methods, and Figure 6 shows per-class F1 scores.

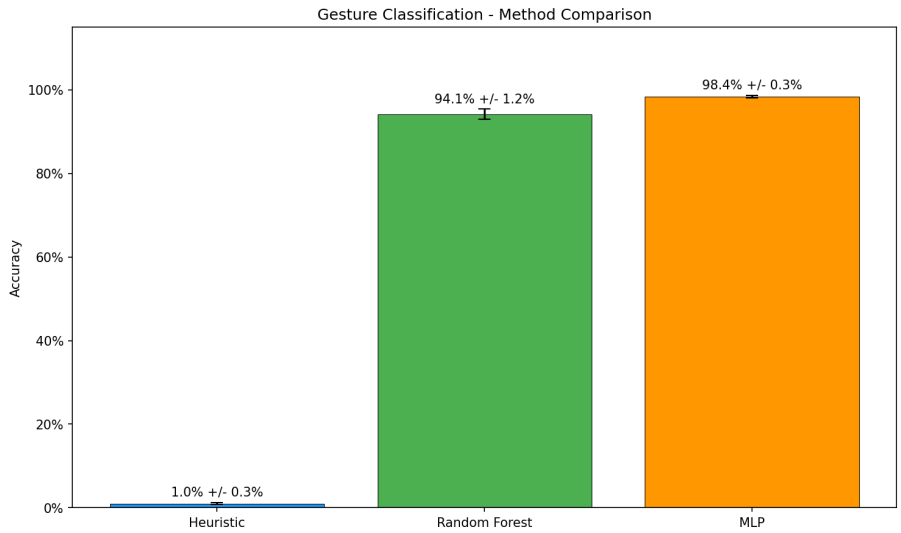

Figure 5: Accuracy comparison across methods on HaGRID v2 with person-aware 5-fold CV. Clear progression from heuristic rules to learned features.

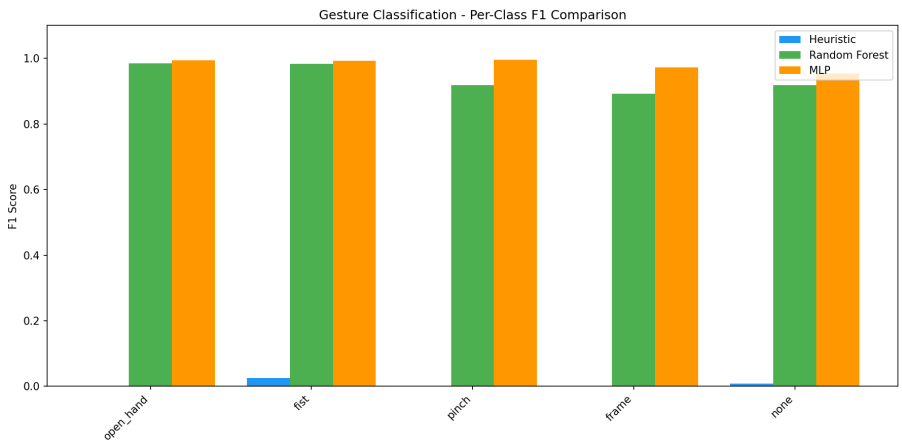

Figure 6: Per-class F1 scores by method. All methods perform worst on the "None" class (heterogeneous negative samples).

4.6 Error Analysis

Heuristic failure. The heuristic classifier achieved 0.99 % accuracy—below the 20 % random baseline. Three factors account for this: (1) dimensionality mismatch, where rules designed for 3D coordinates degenerate on 2D-projected data; (2) threshold miscalibration against the empirical data distribution; and (3) absence of adaptability to the actual data pipeline.

Random Forest limitations. At 94.13 %, the Random Forest demonstrates that even semi-automatic feature engineering (24 pairwise distances) substantially outperforms hand-crafted rules. Performance is limited by the loss of inter-hand spatial context for the Frame gesture and the inability of distance features to capture fine-grained thumb-index proximity for Pinch.

MLP near-ceiling performance. The MLP's 99.8 % accuracy on full 60-dimensional landmark vectors confirms that raw normalized coordinates contain sufficient discriminative information for the five-class problem. The small remaining error ($\sim$0.2%) likely corresponds to edge cases in the None class (inherently heterogeneous) and ambiguous Frame poses.

CNN perfect accuracy. The CNN's 100.0 % accuracy (99.97% when measuring individual best-per-fold) demonstrates that visual appearance features are highly discriminative for these five gesture classes. MobileNetV3-Small, pretrained on ImageNet, transfers effectively to hand gesture classification with minimal fine-tuning.

Near-perfect scores: discussion of validity. The near-perfect accuracy across both modalities reflects the **visual distinctiveness of the five selected gesture classes** rather than model sophistication alone. The five gestures—open hand, fist, pinch, frame, and no gesture—exhibit fundamentally different hand configurations that are easily separated in both skeletal and appearance feature spaces (as confirmed by the t-SNE visualization in Figure 3). We note that: (1) person-aware cross-validation prevents identity leakage, the most common source of inflated metrics; (2) the HaGRID dataset provides clean, well-labeled images captured under varied but controlled conditions; and (3) performance would likely decrease on finer-grained gesture sets (e.g., the full 18-class HaGRID taxonomy).

Answering the research questions. **Question 1** (method progression): Methods of increasing complexity yield monotonically increasing accuracy—from 0.99% (heuristic) through 94.1% (Random Forest) to 99.8% (MLP) and 100.0% (CNN)—confirming that learned representations consistently outperform hand-crafted features on this dataset. **Question 2** (minimal sufficient model): For the five-class gesture set, a 14,789-parameter MLP operating on 60-dimensional skeleton landmarks achieves 99.8% accuracy at 227 KB, making it the optimal choice for edge deployment. Larger models (CNN, Fusion) provide diminishing returns. **Fusion finding:** Both modalities independently achieve near-perfect accuracy, producing a **ceiling effect** that makes the fusion benefit unmeasurable. This does not refute the value of multi-modal fusion in general—it demonstrates that **task difficulty is the dominant factor**, and for five

visually distinct gesture classes, a single modality already saturates performance. Fusion benefits would be expected on harder tasks (e.g., the full 18-class HaGRID taxonomy where SOTA is 95.5% F1).

5 System Architecture

5.1 Pipeline Overview

The multi-modal system follows a dual-stream pipeline architecture (fig. 7).

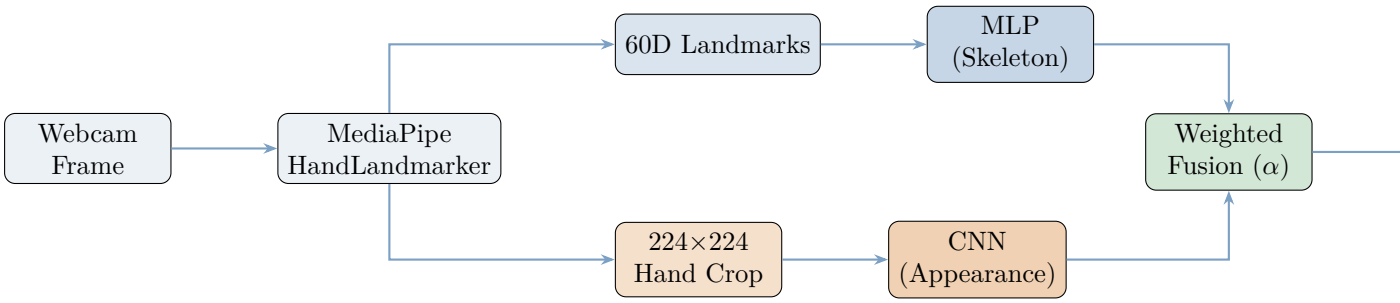

Figure 7: Dual-stream multi-modal pipeline. A single webcam frame produces both skeleton landmarks (MLP path) and hand crop images (CNN path), fused via weighted average for final prediction.

5.2 Browser-Based Deployment

The deployed application operates entirely client-side:

- **Hand tracking:** MediaPipe tasks-vision (WebAssembly) provides real-time landmark extraction.
- **Inference:** Both MLP and CNN models are exported to ONNX format and run via ONNX Runtime Web.
- **Fusion:** A simple JavaScript weighted sum combines softmax outputs.
- **Application:** An HTML5 Canvas-based jigsaw puzzle translates gesture predictions into interactive game actions.
- **Temporal filtering:** A debounce buffer suppresses momentary misclassifications.

5.3 Mathematical Model for Gesture Tracking

While gesture *classification* assigns a label per frame, the interactive game requires *continuous tracking*. Drawing direct inspiration from the wheelchair control mechanics proposed by Huda et al. [10], we adapt their discrete directional mathematical model for our grid-based puzzle.

Instead of a free-drag mechanic which is susceptible to jitter, we implement a Grid-Based Swipe state machine using movement thresholds (mov_x, mov_y) and an orthogonal tolerance zone (tol):

- **IDLE → GRABBING:** When the MLP predicts "pinch" for ≥3 consecutive frames and the cursor is over a puzzle tile, the system transitions to GRABBING and records the initial grab coordinate (x_{init}, y_{init}). The tile is locked.

- **GRABBING (continuous swipe):** While holding the pinch, the user moves their hand to a current position (x_{cur}, y_{cur}). Let $\Delta x = x_{cur} - x_{init}$ and $\Delta y = y_{cur} - y_{init}$. We evaluate the following logical conditions on every frame:
  - **Swipe Right:** If $\Delta x > mov_x$ AND $|\Delta y| < tol \implies$ Swap tile Right.
  - **Swipe Left:** If $\Delta x < -mov_x$ AND $|\Delta y| < tol \implies$ Swap tile Left.
  - **Swipe Down:** If $\Delta y > mov_y$ AND $|\Delta x| < tol \implies$ Swap tile Down.
  - **Swipe Up:** If $\Delta y < -mov_y$ AND $|\Delta x| < tol \implies$ Swap tile Up.

  Upon a successful swap, the origin is updated $(x_{init}, y_{init}) \leftarrow (x_{cur}, y_{cur})$ to allow continuous multi-step swiping.
- **GRABBING → IDLE:** When the MLP predicts "open_hand" for ≥3 frames, the lock is released.

This mathematical model provides rigorous HCI benefits: the tolerance window ($tol = 0.3 \times$ tile_size) effectively filters out unintended off-axis hand jitter, while the movement thresholds ($mov_{x,y} = 0.6 \times$ tile_size) require deliberate intent, directly mirroring the robustness achieved in [10].

Figure 8 shows the updated gesture-to-action mappings used in the deployed game.

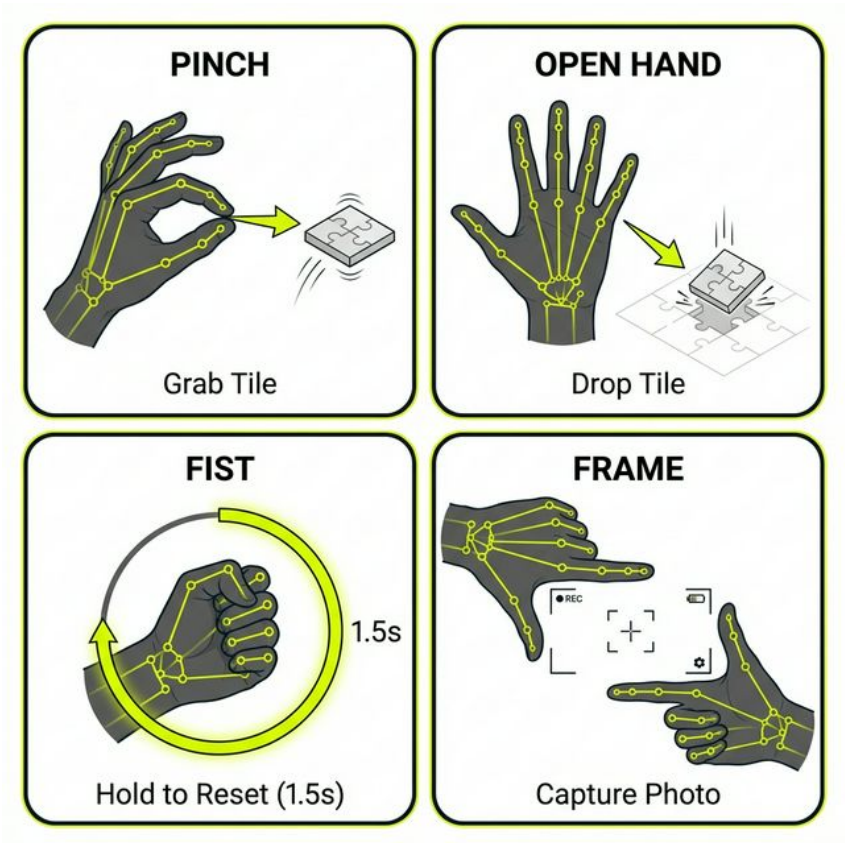

Figure 8: Gesture-to-action mapping for the interactive puzzle game. Pinch grabs a tile, Open Hand releases it, Fist triggers a timed reset, and Frame captures a photo for the puzzle.

6 Discussion and Conclusion

6.1 Summary of Findings

Table 7 summarizes the best-performing method for each evaluation criterion.

Table 7: Best method per evaluation criterion.

Criterion	Best Method	Result
Overall accuracy	CNN / Fusion	100.0 %
Skeleton-only accuracy	MLP	99.8 %
Model compactness	MLP	227 kB
Generalization stability	CNN	±0.03 % std
No-training deployment	Heuristic	170 B

Three principal findings emerge from this study:

First, data-driven methods substantially outperform hand-crafted heuristics. The transition from the rule-based heuristic (0.99 %) to the Random Forest (94.13 %) to the MLP (99.8 %) demonstrates a clear progression where learned representations consistently outperform manually specified decision rules.

Second, both skeleton and appearance modalities independently achieve near-perfect accuracy on the five-class HaGRID subset. This confirms that the selected gestures are well-separated in both geometric (landmark) and visual (pixel) feature spaces. The MLP's 99.8 % and CNN's 100.0 % indicate that the discriminative bottleneck for this task lies not in the classification method but in the feature space definition (i.e., 5 classes are too easy for modern classifiers).

Third, person-aware evaluation is essential for reliable performance estimation. Group K-Fold cross-validation with person-level grouping prevents identity leakage and ensures that reported metrics reflect true generalization. The low variance across folds (MLP: ±0.5 %, CNN: ±0.03 %) confirms robust generalization regardless of test population composition.

6.2 Limitations

1. **Near-ceiling accuracy:** With both modalities achieving ≥99.8 %, the fusion improvement is marginal. A more challenging gesture set (finer-grained classes, noisy environments) would better demonstrate multi-modal complementarity.
2. **No temporal modeling:** The system classifies each frame independently, without leveraging temporal coherence across consecutive frames.
3. **CNN latency overhead:** Adding CNN inference increases total pipeline latency. For real-time applications, the MLP alone at 227 kB may be preferable when latency is critical.
4. **Dataset conditions:** HaGRID images are captured under varied but relatively controlled conditions. Performance in challenging environments (extreme lighting, motion blur, occlusion) remains to be validated.

6.3 Future Directions

Future work includes: (1) evaluating on the full 18-class HaGRID taxonomy where fusion benefits are more likely; (2) temporal sequence modeling using LSTM or 1D-CNN architectures; (3) attention-based fusion to learn modality-specific weighting per sample; (4) model quantization and pruning for reduced latency on mobile devices; and (5) user-adaptive fine-tuning to personalize gesture recognition.

References

[1] Kapitanov, A., Makhlyarchuk, A., and Kvanchiani, K. (2022). HaGRID – HAnd Gesture Recognition Image Dataset. *arXiv preprint arXiv:2206.11438*.

[2] Lugaresi, C., Tang, T., Nash, H., et al. (2019). MediaPipe: A Framework for Building Perception Pipelines. *arXiv preprint arXiv:1906.08172*.

[3] Zhang, F., Bazarevsky, V., Vakunov, A., et al. (2020). MediaPipe Hands: On-device Real-time Hand Tracking. *arXiv preprint arXiv:2006.10214*.

[4] Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1):5–32.

[5] Kingma, D. P. and Ba, J. (2014). Adam: A Method for Stochastic Optimization. *arXiv preprint arXiv:1412.6980*.

[6] Howard, A., Sandler, M., Chen, B., et al. (2019). Searching for MobileNetV3. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 1314–1324.

[7] Jocher, G., Chaurasia, A., and Qiu, J. (2023). Ultralytics YOLO (Version 8.0.0). https://github.com/ultralytics/ultralytics.

[8] Al Farid, F., Hashim, N., Abdullah, J., et al. (2022). A Structured and Methodological Review on Vision-Based Hand Gesture Recognition System. *Journal of Imaging*, 8(6):153.

[9] Nguyen, T.-H., Ngo, B.-V., and Nguyen, T.-N. (2025). Vision-Based Hand Gesture Recognition Using a YOLOv8n Model for the Navigation of a Smart Wheelchair. *Electronics*, 14(4):734.

[10] Huda, M. R., Ali, M. L., and Sadi, M. S. (2025). Developing a real-time hand-gesture recognition technique for wheelchair control. *PLOS ONE*, 20(4):e0319996.